How Accurate Is Regression-Based Global Effect Estimation Under Network Interference? Error Anatomy, Design Diagnostics, and Failure Modes

Anonymous Authors[1]

Abstract

Regression-based estimators of the global average treatment effect (GATE) under network interference fit an outcome model on the experimental data and predict "all treated" versus "all control". How accurate are they, what determines their error, and when do they fail? For least-squares estimators we show an exact decomposition: the estimator is a weighted sum $\sum_i w_i y_i$ whose weights depend only on the realised assignment, so the noise error is exactly $\sigma\|w\|_2$ and the misspecification error is controlled by $\|w\|_1$ and the approximation residual. For Bernoulli designs we derive that the noise error grows like $\sqrt{d/n}$ in the degree d, and that the large-degree bias equals the local slope of the exposure–response curve at the treatment probability rather than its secant slope. In 11 simulation experiments (eight interference mechanisms; synthetic graphs, ego-Facebook and Facebook100-Stanford3; Bernoulli, cluster and saturation designs; eleven estimators) the noise prediction holds to within a few percent (ratio 1.00) and the bias prediction matches the observed bias within Monte-Carlo error in 30 settings, without fitted parameters. The bias does not shrink with n; it changes sign with the treatment probability. Cluster randomisation removes most of the extrapolation error only when the graph has small-cut clusters, and then ignoring interference is nearly as good; flexible learners (gradient boosting) and richer feature sets were worse than a two-feature linear regression; regression can be worse than the naive difference in means at high degree or weak interference; edge errors in the analyst's graph bias Bernoulli-design estimates roughly proportionally; and i.i.d./robust standard errors under-cover (estimated SEs as small as 0.32 of the true SD) whereas cluster-robust errors help only when the clusters are meaningful. All conclusions are conditional on our synthetic mechanisms, and we report failures and limitations.

1. Introduction

Online experiments on social networks and marketplaces violate the no-interference assumption: a user's outcome depends on the treatments of the users she is connected to. The quantity a platform usually wants is the *global average treatment effect* (GATE), the mean outcome difference between "everyone treated" and "nobody treated". Because this is a counterfactual that is never observed in a single experiment, every estimator must extrapolate. The difference-in-means (DM) estimator extrapolates not at all, and is biased by the spillover it ignores. Design-based estimators (Horvitz–Thompson/Hájek on exposure conditions (Aronow & Samii, 2017; Ugander et al., 2013)) are unbiased for a coarsened target but have large variance when full-neighbourhood exposure is rare. A third family, *regression-based* estimators (Chin, 2019; Han & Ugander, 2023), fits a model from the assignment (and graph-derived exposure features) to the outcome, and then predicts the outcome under all-treated and all-control assignments. These are attractive because they are cheap, use all units, and need no exposure probabilities. But the prediction points "all treated" and "all control" usually lie far outside the observed feature distribution, and it is not well understood *how accurate* such estimators are, *what determines* their error, and *when they fail*.

We study this question with a controlled simulation study backed by an exact error decomposition. Our contributions are:

1. **Exact decomposition and design diagnostics (§3).** For least-squares GATE estimators we show $\hat{\tau} = \sum_i w_i y_i$ with weights w computable from the realised assignment alone. The error splits exactly into a noise term with variance $\sigma^2\|w\|_2^2$ and a misspecification/extrapolation term controlled by $\|w\|_1$ and the approximation residual. On regular graphs under Bernoulli(p) designs,

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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$n\|w\|_2^2 \to (d+1)/(p(1-p))$, so error grows like $\sqrt{d/n}$, and the large-degree bias equals the *local slope* $g'(p)$ of the exposure–response curve rather than its secant slope.

2. **A large, reproducible simulation study (§4)** over eight interference mechanisms (linear, saturating, convex, threshold, two-hop, heterogeneous, equilibrium peer effects), six graphs (ER, BA, WS, SBM, and the public ego-Facebook and Facebook100-Stanford3 graphs), Bernoulli, cluster and randomised-saturation designs, with eleven estimators including a gradient-boosted learner and an Hájek baseline. All results report Monte-Carlo uncertainty.

3. **Findings, positive and negative.** The noise term is predicted almost exactly; the bias is *not* reduced by more data; cluster randomisation helps only when the graph has good clusters; flexible learners are worse than linear regression under Bernoulli designs because of extrapolation; richer linear models overfit; neighbourhood misspecification and standard error estimators are examined, with failures reported (§5).

2. Related work

Design-based estimation under interference. Aronow & Samii (2017) define exposure mappings and give Horvitz–Thompson (HT) estimators; Ugander et al. (2013) propose graph cluster randomisation to make full-neighbourhood exposure likely, and Eckles et al. (2017) study its bias–variance trade-off and analysis methods using neighbour treatment information; Karrer et al. (2021) describe a deployed system. Leung (2022) derives rate-optimal cluster designs for HT estimation of the GATE under spatial interference, and Sävje et al. (2021) characterise what standard estimators estimate under unknown interference. Basse & Airoldi (2018) show limits of design-based inference under arbitrary interference. Cortez et al. (2022) give staggered-rollout designs without network knowledge, and Yu et al. (2022) estimate the total treatment effect with unknown network structure.

Regression-based estimation. Chin (2019) proposes regression adjustment that predicts all-treated and all-control outcomes, with properties under linear models; Han & Ugander (2023) select graph-based and treatment-based covariates and use block bootstrap; Li & Wager (2022) analyses estimators under random-graph asymptotics. Baird et al. (2018) study randomised saturation designs, a related design idea we include. Chen et al. (2023) optimise the assignment covariance for HT estimators in a one-hop model.

Positioning. Prior regression work establishes validity under a correctly specified model or provides methodology. We instead ask how large the error is, and *what it is made of*, across mechanisms and graphs, give an exact weight representation with post-assignment diagnostics, and map when linear and flexible regressions fail (extrapolation, high degree, poor clusters, wrong graph). Graph neural networks (Kipf & Welling, 2017; Defferrard et al., 2016) are an alternative for learning exposure features; we do not evaluate them (§6).

3. Setting and error anatomy of regression GATE estimators

Setting. A fixed, connected, undirected graph with adjacency A has n nodes with degrees d_i. A design draws $z \in \{0,1\}^n$. Unit i has expected outcome $\mu_i(z)$ and we observe $y_i = \mu_i(z)+\varepsilon_i$ with independent noise, $\mathrm{Var}(\varepsilon_i) = \sigma^2$. The estimand is $\tau = \frac{1}{n}\sum_i[\mu_i(\mathbf{1}) - \mu_i(\mathbf{0})]$ (GATE). The population (graph, $\mu_i(\cdot)$) is fixed; randomness is in z and ε.

Regression estimator. Let $\phi_i(z) \in \mathbb{R}^q$ be features built from the assignment and the (observed) graph, e.g. $\phi_i = (z_i, x_i)$ with $x_i = \frac{1}{d_i}\sum_j A_{ij}z_j$ the treated neighbour fraction. Fit y on $(1, \phi_i(z))$ by least squares to get $\hat\beta$, and set $\hat\tau = \frac{1}{n}\sum_i[\phi_i(\mathbf{1}) - \phi_i(\mathbf{0})]^\top\hat\beta_\phi$. With $X = [\mathbf{1}, \Phi(z)]$ and $\tilde c = (0, \frac{1}{n}\sum_i(\phi_i(\mathbf{1}) - \phi_i(\mathbf{0})))$:

$$\hat\tau = \tilde c^\top(X^\top X)^{-1}X^\top y = \textstyle\sum_i w_i y_i, \qquad w = X(X^\top X)^{-1}\tilde c. \tag{1}$$

Proposition 3.1 (Exact error decomposition)**.** *w depends only on z and the graph, $\sum_i w_i = 0$ and $X^\top w = \tilde c$. For* any *β, writing $\mu_i(z') = \phi_i(z')^\top\beta_\phi + \beta_0 + r_i(z')$,*

$$\hat\tau - \tau = \underbrace{\textstyle\sum_i w_i\varepsilon_i}_{\text{noise}} + \underbrace{\textstyle\sum_i w_i r_i(z) - \frac{1}{n}\sum_i\big[r_i(\mathbf{1}) - r_i(\mathbf{0})\big]}_{\text{misspecification / extrapolation}}.$$

Given z, the noise term has mean 0 and variance $\sigma^2\|w\|_2^2 = \sigma^2\tilde c^\top(X^\top X)^{-1}\tilde c$ (exactly, for any n, no asymptotics). The second term vanishes if μ is in the model, and otherwise satisfies $|\cdot| \le \|w\|_1\|r(z)\|_\infty + |\frac{1}{n}\sum_i(r_i(\mathbf{1}) - r_i(\mathbf{0}))|$ and $|\sum_i w_i r_i| \le \|w\|_2\|r(z)\|_2$.

Proof. Since $X^\top w = \tilde c$, $\sum_i w_i\mu_i(z) = \tilde c^\top\beta + \sum_i w_i r_i(z)$, while $\tau = \tilde c^\top\beta + \frac{1}{n}\sum_i(r_i(\mathbf{1}) - r_i(\mathbf{0}))$. Subtract and add $\sum_i w_i\varepsilon_i$. □

So the realised assignment fixes two numbers that can be computed *before any outcome is seen*: $\|w\|_2$ (sensitivity to noise) and $\|w\|_1$ or $\|w\|_2$ (sensitivity to misspecification). We use them as design diagnostics and test them in §5. In the experiments the second term is measured directly by refitting on the noiseless outcomes $\mu(z)$: this gives the *assignment-induced* part $S = \sum_i w_i\mu_i(z) - \tau$ (random only through z), whose mean is the bias and whose variance is the design variance. The total error is $S + N$ with $N = \sum_i w_i\varepsilon_i$.

Proposition 3.2 (Bernoulli design, regular graph; large n)**.** *Let A be d-regular with bounded d, $z_i \sim \mathrm{Bern}(p)$ i.i.d., $\phi = (z, x)$. Then $n\|w\|_2^2 \to \frac{d+1}{p(1-p)}$ as $n \to \infty$. Hence $\mathrm{RMSE}_{\text{noise}} \approx \sigma\sqrt{(d+1)/(np(1-p))}$: error grows like*

$\sqrt{d}$ *although each unit has more neighbours. If* $\mu_i(z) = a_i + \tau_d z_i + \gamma g(x_i)$ *for a smooth* g*, and the* a_i *are exogenous baseline terms, the population least-squares slope on* x *is* $\beta_x = \mathrm{Cov}(g(x), x)/\mathrm{Var}(x) = g'(p) + \frac{(1-2p)g''(p)}{2d} + O(d^{-2})$*, so the large-*$n$ *bias is*

$$\text{bias} = \gamma\big[\beta_x - (g(1)-g(0))\big] \xrightarrow{d\to\infty} \gamma\big[g'(p) - (g(1)-g(0))\big].$$

Sketch. In a d-regular graph without self-loops, z_i and x_i are uncorrelated with $\mathrm{Var}(z) = p(1-p)$, $\mathrm{Var}(x) = p(1-p)/d$, and bounded dependence gives convergence of $X^\top X/n$, so $\tilde{c}^\top (X^\top X/n)^{-1}\tilde{c} = (1,1)\,\mathrm{diag}(v, v/d)^{-1}(1,1)^\top = (1+d)/v$. For the slope, $x = B/d$ with $B \sim \mathrm{Bin}(d, p)$; expand g around p and use $E(x-p)^3 = p(1-p)(1-2p)/d^2$. The direct effect τ_d is recovered since $z \perp x$. □

Reading. Under Bernoulli designs with many neighbours, nearly all units see $x_i \approx p$, so the regression identifies only the *local* exposure–response slope at p; the GATE needs the secant slope from 0 to 1. They agree iff g is (locally) linear, or by coincidence (quadratic g at $p = \frac{1}{2}$: tangent equals secant). Cluster randomisation spreads x_i towards 0 and 1 (units in treated clusters see $x_i \approx 1$), so the regression interpolates instead of extrapolating, at the price of correlation between z and x (larger $\|w\|$ if clusters are poor) and of fewer independent randomisation units. We test all of this empirically, including the proposition, in §5.

4. Experimental design

Graphs. Synthetic (n=3000 unless stated; largest component): Erdős–Rényi (ER, mean degree 16), Barabási–Albert (BA, m=8), Watts–Strogatz (WS, k=16, β=0.1), stochastic block model (SBM, 30 blocks, mean degree 16, fraction of between-block edges `mix` default 0.1); public graphs: SNAP ego-Facebook (McAuley & Leskovec, 2012) (4039 nodes, mean degree 43.7) and Facebook100 Stanford3 (Traud et al., 2012; Rossi & Ahmed, 2015) (11,586 nodes, mean degree 98.1).

Outcome model. $\mu_i(z) = a_i + \tau_d z_i + \gamma\, g_i(z)$ with τ_d=γ=1 and σ=1 by default. Baselines $a_i = 1 + 0.5\,\mathrm{zlogdeg}_i + 0.5\,b_{c(i)} + 0.7\,\nu_i$ (b_c: Louvain-community effect, ν_i i.i.d. $N(0,1)$) make outcomes homophilous and degree-dependent and are *not* fully known to the regression. Writing $x_i = \frac{1}{d_i}\sum_j A_{ij} z_j$, the eight mechanisms differ in g, normalised so that $g(\mathbf{1}) - g(\mathbf{0}) = 1$ (GATE= 2, so relative errors are comparable): `linfrac`: x_i; `lincount`: $\sum_j A_{ij} z_j / \bar{d}$ (hub amplification); `sat`: $(1 - e^{-4x_i})/(1 - e^{-4})$ (diminishing returns); `quad`: x_i^2; `thresh`: logistic in x_i (centre 0.7, slope 10, rescaled; complex contagion); `twohop`: $\frac{1}{2}x_i + \frac{1}{2}(Wx)_i$; `hetero`: $(1 + 0.5u_i)x_i$ with latent u_i partly correlated with degree; `peer`: equilibrium peer effects $\mu = (I - \rho W)^{-1}(a + \tau_d z)$, ρ=0.5, W row-normalised (interference of unbounded range). The GATE of every population is computed exactly from $\mu(\mathbf{1}) - \mu(\mathbf{0})$.

Designs. *Bernoulli*(p), default p=0.5; *cluster*: Louvain clusters (Blondel et al., 2008) (K clusters, fraction cut edges reported), half of the clusters treated by complete randomisation; *randomised saturation* (clusters draw $p_c \in \{.1, .3, .5, .7, .9\}$, units Bernoulli($p_c$)).

Estimators. **DM**; OLS on (z, x) (**OLS-x**, the model of Prop. 3.2), plus degree (**xd**), plus x^2 and zx (**quad**), plus two-hop exposure (**x2**), a 10-feature polynomial/interaction model (**rich**), count exposure without/with degree control (**c**, **cd**); **GBDT**: histogram gradient boosting on $(z, x, \text{count}, \text{2-hop}, \text{degree})$ predicting at all-1/all-0 features; **Hájek**: exposure-conditional Hájek estimator for "treated with $x_i \geq 0.75$" vs. "control with $x_i \leq 0.25$" with exposure probabilities obtained by Monte-Carlo (M=400 design draws); and an **oracle** OLS using the true exposure basis (plus degree). Hájek targets a coarsened (q=0.75) contrast and so is biased for the GATE by construction.

Metrics and uncertainty. Per configuration we run R=100–500 independent assignments and noise draws (fixed population) and report *bias*, *SD*, and *RMSE* (primary: it is the quantity a practitioner experiences, and it penalises bias and variance), mostly as relative RMSE = RMSE/$|\tau|$; 95% bootstrap CIs over replicates for RMSE and Monte-Carlo standard errors for bias. For each replicate we also refit on noiseless $\mu(z)$ to separate S (assignment-induced; its mean is the *bias*) from N (noise) as in Prop. 3.1. Coverage of 95% intervals is evaluated in §5.

5. Results

Unless noted, n=3000, σ=1, γ=1 (GATE = 2), error bars/shaded values are 95% bootstrap CIs over replicates, and "bias" is the Monte-Carlo mean error (its standard error is reported where needed). All numbers come from the logged runs in `results/` (reproduction map in Appendix A).

R1: Overall accuracy (Table 1, Fig. 7 in the appendix). Ignoring interference (DM) under Bernoulli designs has relative RMSE ≈ 0.50 in all 48 population×mechanism cases: the bias is the entire spillover ($-\gamma$) and does not shrink with n. Regression estimators with an exposure feature cut this to a median of 0.16–0.32 (OLS-x) per graph under Bernoulli designs, with the best case (oracle features) at 0.09–0.20 but a worst case up to 0.95 (Table 1). Under cluster randomisation the answer depends on the graph: on graphs with a small edge cut (WS, SBM, ego-Facebook: cut ≤ 0.16) DM itself drops to 0.12–0.17 and regression gives 0.10–0.19, i.e. regression helps little over DM there (on ego-Facebook DM is the best of the six methods). On graphs with poor clusters (ER/BA, cut 0.82; Stanford3, cut 0.64) cluster randomisation brings no gain: ER and BA

Louvain clusters (K=110) leave 82% of edges cut, and Stanford3's 57 clusters give a large cluster-level variance (OLS-x SD $\approx$ 0.5–0.6 vs. $\approx$ 0.15–0.18 under Bernoulli, excluding `peer`).

R2: The noise term is predicted exactly; error grows with degree (Fig. 1). Across all 2,010 (configuration, OLS estimator) pairs of Experiments 1–6 and 8–10 (`fig_diag.py`) the empirical SD of $\sum_i w_i\varepsilon_i$ divided by $\sigma\|w\|_2$ has median 1.001 (5–95% range 0.92–1.09; log-log correlation 0.999), as Prop. 3.1 implies. With $\|w\|_2^2 \approx (\bar{d}{+}1)/(np(1{-}p))$, RMSE of a *correctly specified* regression grows from 0.10 ($\bar{d}$=4) to 0.56 ($\bar{d}$=128) at n=3000 (OLS-x, no community effect) (Fig. 1a), and falls like $n^{-1/2}$ (Fig. 1c); on these (irregular, Poisson-degree) ER graphs $n\|w\|_2^2$ divided by $(\bar{d}{+}1)/(p(1{-}p))$ is 0.79 at $\bar{d}$=4, 0.95 at $\bar{d}$=16, 1.04 at $\bar{d}$=128, so Prop. 3.2 is accurate for moderate/large degree and optimistic for sparse graphs with heterogeneous degrees; $\|w\|_1$ grows like $\sqrt{\bar{d}}$ and is constant in n (6.4 at $\bar{d}$=16 for every n from 500 to 16,000). The RMSE is larger than the noise term alone, because outcome heterogeneity that the regression does not model (unit baselines and community effects) is projected through the same weights: with community effects the assignment-induced SD ("sdS") exceeds the pure-noise SD (e.g. 0.22 vs. 0.14 at $\bar{d}$=16). Consequently RMSE is barely sensitive to σ in our baseline (Appendix Fig. 10): doubling σ from 1 to 2 changes OLS-x RMSE from 0.39 to 0.44 (linear case).

R3: Bias is the slope-at-p error and does not vanish with n (Fig. 2, 3; Prop. 3.2). For the two curved mechanisms, OLS-x bias has the sign and size predicted by $g'(p) - (g(1) - g(0))$: in 30 (graph, mechanism, p) cases of Experiment 9 the empirical noiseless bias and the parameter-free prediction have correlation 1.00, mean absolute difference 0.021 and maximum 0.07, with every difference within 1.7 Monte-Carlo standard errors (Table 5 in the appendix). The bias flips sign: for `sat` it is $+0.86$ at p=0.2 and -0.37 at p=0.5, and for `thresh` it is -0.83 at p=0.2 and $+0.25$ at p=0.5 (SBM, Exp. 9). With $p \le 0.1$ the linear model's RMSE is 1.7–2.2 for `sat` (GATE = 2). At p=0.5 the quadratic x^2 is estimated without bias, purely because tangent and secant slopes coincide. The bias does not decrease with data: for `sat` under Bernoulli designs OLS-x bias is $-0.33, -0.38, -0.41, -0.37, -0.39$ for n = 1k, 2k, 4k, 8k, 16k, and its RMSE plateaus at 0.42 while the oracle model's RMSE keeps falling to 0.22 (Appendix Fig. 9). With fixed n=3000, higher degree makes things worse (Fig. 3): OLS-x RMSE (Bernoulli, `sat`) rises from 0.17 ($\bar{d}$=4) to 2.9 ($\bar{d}$=128), exceeding DM (1.00) between $\bar{d}$=32 (0.86) and $\bar{d}$=64 (1.38); the noise term scales as $\sqrt{d/n}$, so the relevant quantity is d/n.

R4: Design matters mostly through cluster quality (Fig. 4, Table 2). Cluster randomisation removes most of the extrapolation: on SBM with 10% between-block edges, `sat` OLS-x has bias $+0.01$ (vs. -0.39 for Bernoulli(0.5)) and $\|w\|_1 = 2.4$ (vs. 6.4); absolute RMSE (GATE = 2) falls from 0.55 to 0.22, and DM's from 1.0 to 0.26. Its advantage erodes as clusters get worse: at 60% between-block edges the cluster design's OLS-x RMSE (0.50) is worse than Bernoulli(0.5) (0.43), and randomised saturation lies between Bernoulli and cluster randomisation for $\le$ 30% between-block edges (OLS-x RMSE 0.40–0.49) and is worst at 60% (0.82); it was never better than plain cluster randomisation here. A small p (0.2) is clearly harmful for regression (bias $+0.88$ for `sat`), and under Bernoulli(0.2) the Hájek estimator found no treated-and-saturated and control-and-isolated units in 98% of replicates (undefined); for `thresh` even the *correctly specified* model has RMSE 1.4–2.1 at p=0.2 because the design never excites the steep region (large $\|w\|$): correct specification is not sufficient. The cluster design retains a variance floor from the few randomisation units (K=15–30): its RMSE does not fall with degree, so it is better than Bernoulli at high degree and worse at low degree (Fig. 3; for OLS-x the crossover is near $\bar{d} \approx$ 8–16 with 15 blocks).

R4b: Regression tolerates finer clusters than DM does (Appendix Table 6). When the cluster resolution is increased (more, smaller clusters; higher cut) DM degrades monotonically on ego-Facebook (RMSE 0.34 $\to$ 0.56 from K=16 to 672, `sat`), whereas OLS-x is non-monotone but ends lower: 0.34 (K=16, cut 0.04), 0.41 (K=27), 0.26 (K=238, cut 0.53); similarly on Stanford3 (0.60 $\to$ 0.41 from K=10 to 197) and WS (0.24 $\to$ 0.17, K=27 $\to$ 156). It does not help indefinitely: on SBM with 30% between-block edges OLS-x worsens (0.23 $\to$ 0.49) when Louvain splits the blocks (K=376, cut 0.78). The right granularity trades the variance floor from few randomisation units against the cut and differs between DM and regression.

R4c: Outcome homophily drives the variance of cluster designs (Appendix Table 7). Our default community-level baseline effect (sd 0.5) is one choice. Varying it from 0 to 2 (SBM, `sat`) changes OLS-x RMSE under cluster randomisation from 0.06 to 0.84 (and under Bernoulli from 0.41 to 1.37), while DM under Bernoulli stays at 1.00 (pure bias). With strong community-level baselines, Bernoulli-design OLS-x (1.37) is worse than DM (1.00), and the accuracy advantage of the cluster design over DM under cluster randomisation shrinks (0.84 vs. 0.87). The headline accuracy numbers (R1) therefore depend on how much baseline outcome variation is shared within clusters, a quantity the analyst can estimate from pre-experiment data but we did not exploit.

R5: Flexible learners and richer features do not fix extrapolation (Table 1, Appendix Fig. 7). Gradient boosting has median relative RMSE 0.23–0.44 under Bernoulli de-

Table 1. Relative RMSE (RMSE/—GATE—): median over the eight mechanisms (min–max in parentheses); n as listed (3000 synthetic; 4039 egoFB; 11,586 Stanford3), 200 replicates per cell. Bold: lowest median per row. Bern: Bernoulli(0.5); Clust: cluster randomisation with Louvain clusters (K clusters, fraction of edges cut).

Graph	Design	DM	OLS-x	OLS-quad	Oracle	GBDT	Hájek
ER	Bern	0.50 (0.48–0.50)	0.16 (0.13–0.27)	0.15 (0.12–0.27)	**0.12 (0.06–0.56)**	0.40 (0.37–0.46)	0.33 (0.31–0.42)
ER	Clust (K=110, cut 0.82)	0.42 (0.39–0.45)	0.19 (0.16–0.27)	0.17 (0.15–0.26)	**0.15 (0.08–0.60)**	0.35 (0.32–0.44)	0.28 (0.21–0.37)
BA	Bern	0.50 (0.49–0.50)	0.19 (0.12–0.28)	0.19 (0.12–0.28)	**0.12 (0.05–0.55)**	0.37 (0.35–0.43)	0.33 (0.24–0.42)
BA	Clust (K=111, cut 0.82)	0.41 (0.38–0.44)	0.19 (0.13–0.26)	0.19 (0.13–0.26)	**0.12 (0.05–0.61)**	0.31 (0.26–0.41)	0.28 (0.19–0.35)
WS	Bern	0.50 (0.48–0.50)	0.20 (0.18–0.33)	0.20 (0.18–0.33)	**0.18 (0.08–0.36)**	0.42 (0.36–0.61)	0.34 (0.28–0.42)
WS	Clust (K=49, cut 0.16)	0.12 (0.09–0.20)	0.10 (0.09–0.18)	0.10 (0.09–0.18)	**0.10 (0.08–0.16)**	0.34 (0.32–0.57)	0.11 (0.09–0.19)
SBM	Bern	0.50 (0.49–0.50)	0.21 (0.17–0.38)	0.20 (0.17–0.37)	**0.18 (0.07–0.95)**	0.44 (0.41–0.67)	0.33 (0.26–0.47)
SBM	Clust (K=30, cut 0.10)	0.12 (0.11–0.21)	0.11 (0.10–0.20)	**0.10 (0.10–0.20)**	0.10 (0.10–0.20)	0.22 (0.21–0.35)	0.12 (0.11–0.21)
egoFB	Bern	0.50 (0.49–0.50)	0.32 (0.23–0.52)	0.26 (0.22–0.46)	**0.20 (0.09–0.68)**	0.36 (0.33–0.61)	0.36 (0.27–0.51)
egoFB	Clust (K=27, cut 0.07)	**0.17 (0.15–0.29)**	0.19 (0.17–0.35)	0.21 (0.18–0.37)	0.20 (0.17–0.34)	0.52 (0.50–1.08)	0.17 (0.16–0.31)
Stanford3	Bern	0.50 (0.50–0.50)	0.18 (0.08–0.46)	0.17 (0.07–0.46)	**0.09 (0.04–0.59)**	0.23 (0.20–0.32)	0.31 (0.19–0.50)
Stanford3	Clust (K=57, cut 0.64)	0.32 (0.27–0.39)	0.29 (0.25–0.47)	**0.26 (0.24–0.43)**	0.30 (0.16–0.76)	0.27 (0.26–0.44)	0.31 (0.25–0.44)

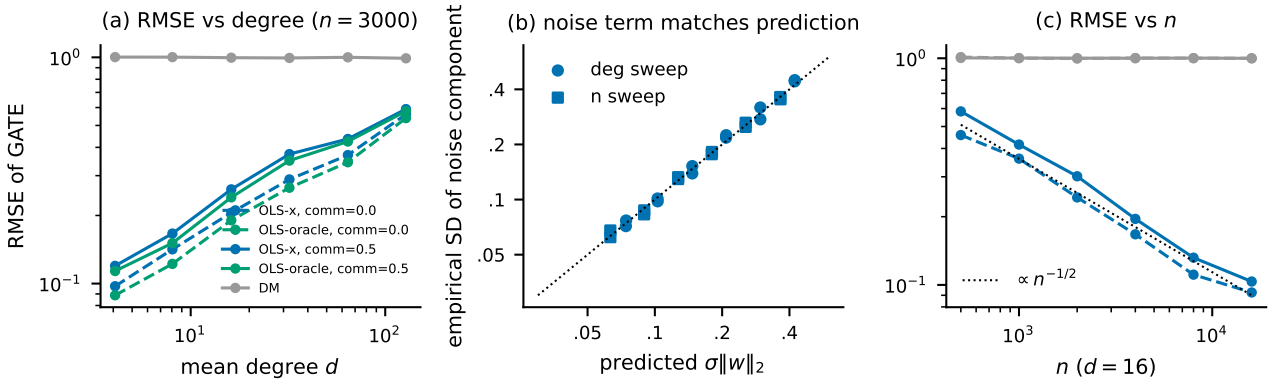


Figure 1. Noise term of the OLS estimator. (a) RMSE vs mean degree on ER graphs under Bernoulli(0.5) with correctly specified linear exposure ("comm": strength of community-level baseline effects; 300 replicates). DM is flat at ≈ 1 (bias $= -\gamma$). (b) Empirical SD of the noise term $\sum_i w_i \varepsilon_i$ (obtained by refitting on noiseless outcomes) vs. the prediction $\sigma\|w\|_2$ (diagonal). (c) RMSE vs. n; the dotted line has slope $-1/2$.

signs (OLS-x: 0.16–0.32), i.e. worse than OLS-x on all six graphs; on the synthetic graphs its bias is towards zero (-0.2 to -0.4 median; tree predictions cannot leave the range of observed exposures) while on the two real graphs it is near zero ($+0.02$) with a large SD, and its SD is 1.2–2.6× that of OLS-x; it does not converge with n in the range tried (RMSE 0.65 at n=16k for `sat`, bias -0.47). The 10-feature polynomial model `rich` is the worst regression (median over cells 0.86 vs. 0.21 for OLS-x): extra features inflate $\|w\|_2$ (mean 0.71 vs. 0.12) more than they remove bias. Adding x^2 and zx (`quad`) is neutral (median 0.21 vs. 0.21), and the two-hop feature removes the bias of the `twohop` mechanism ($+0.07$ vs. -0.47, ER/Bernoulli) but its variance ($\|w\|_2 = 0.36$) makes the RMSE higher than OLS-x in all 12 design×graph cells (1.11 vs. 0.54 on ER/Bernoulli): bias–variance, not a free fix.

R6: Regression only pays off above a spillover threshold (Appendix Fig. 10). With γ=0 DM has RMSE 0.06 against 0.37 for OLS-x (linear mechanism, Bernoulli): the price of the regression is $\approx 6\times$ larger error when there is no interference; DM wins at γ=0 and OLS-x at γ=0.5 (0.34 vs. 0.51), so the break-even lies in between, near $\gamma \approx 0.35$ by linear interpolation (spillover $\approx 26\%$ of the GATE; we did not run intermediate values). Under cluster randomisation on the SBM graph DM and OLS-x are comparable for small γ (0.22 vs. 0.22 at γ=0) and OLS-x is better for $\gamma \geq 2$ (e.g. 0.24 vs. 0.44 for `sat`).

R7: Wrong graph. (Fig. 5.) If the analyst's graph deletes a fraction r of the true edges and adds as many random ones, the Bernoulli-design bias grows roughly like $-r$: OLS-x bias for `linfrac` is $-0.07, -0.24, -0.50, -0.75$ for $r = 0.1, 0.25, 0.5, 0.75$ (RMSE 0.34 → 0.78 for r from 0 to 0.75), approaching DM's. Under cluster randomisation the estimator is robust (-0.01 at r=0.25, -0.08 at r=0.75)

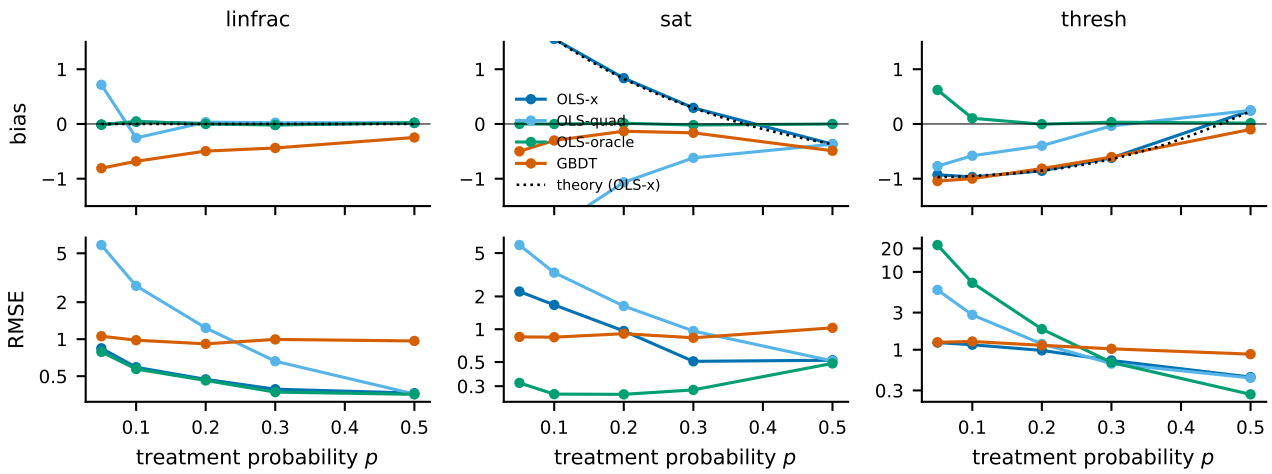


Figure 2. Bias and RMSE vs. Bernoulli treatment probability p on the SBM graph (n=3000, $\bar{d}$=16). Dotted: prediction of the linear-in-x regression from Prop. 3.2 (finite-d binomial version), no fitted parameters. 200 replicates per point.

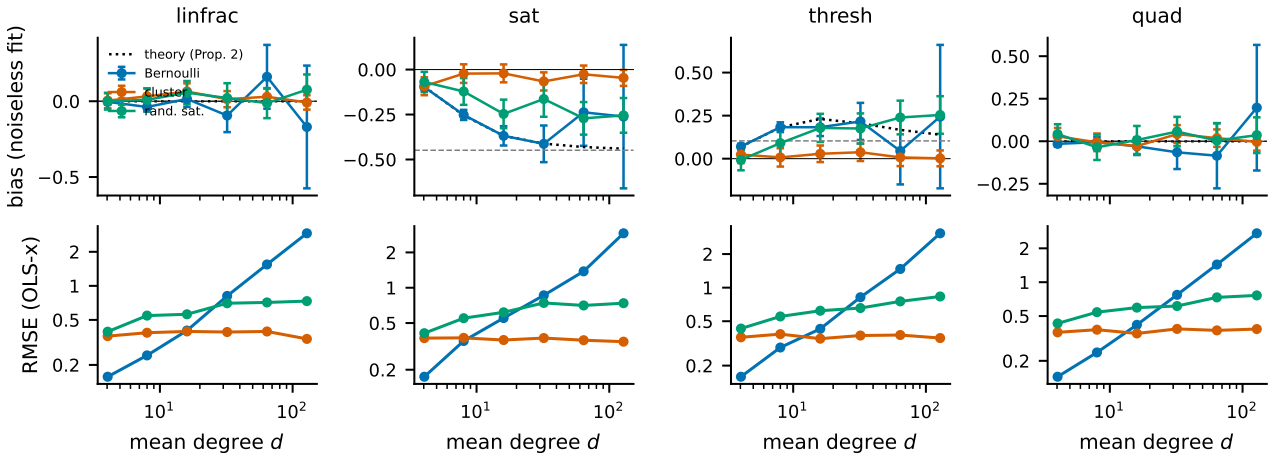


Figure 3. Degree sweep on SBM graphs (n=3000, 15 blocks, 200 replicates). Top: bias of OLS-x from noiseless fits ±1.96 MC s.e.; dotted: Prop. 3.2; dashed grey: $d \to \infty$ limit. Bottom: RMSE.

because x is dominated by own-cluster treatment. Using the correct feature form does not remove this bias when the graph is wrong (oracle ≈ OLS-x for `linfrac`; for `sat` oracle RMSE 0.62 vs. 0.73 at r=0.5).

R8: Uncertainty quantification is the weakest link (Table 3). For OLS-x on 16 populations, 95% intervals from the usual i.i.d. model SE or heteroskedasticity-robust (HC0/HC3) SEs under-cover even when the model is correct: for `linfrac` on the SBM graph coverage is 0.76 (Bernoulli) and 0.48 (cluster) as the SEs are ≈ 0.60 and 0.32× the true SD, because residuals are positively correlated through neighbouring units and shared community effects. SEs clustered on the design clusters (CR) restore near-nominal coverage when the clusters are meaningful (SBM, correct model: 0.93, 0.92) and fail when they are not (ER: 0.89, 0.82). When the model is biased (`sat`, `peer`; Appendix Table 8) coverage of the i.i.d. intervals falls to 0.34–0.61 and even CR coverage is as low as 0.53 (ER, `peer`).

R9: Diagnostics (Fig. 6). $\|w\|_2$ predicts the noise SD almost perfectly (R2), so a practitioner can compute $\hat{\sigma}\|w\|_2$ from the realised assignment. Across all configurations with a correct graph, $\|w\|_1$ rank-correlates with |bias| of OLS-x for curved responses (Spearman 0.81 for `sat`, 0.83 for `thresh`, 0.75 for `twohop`; 112, 80, 30 configurations) and weakly where the model is (nearly) correct (0.21 for `linfrac`, where the true bias is zero and the correlation reflects only Monte-Carlo noise); it is a risk indicator, not a bias estimate.

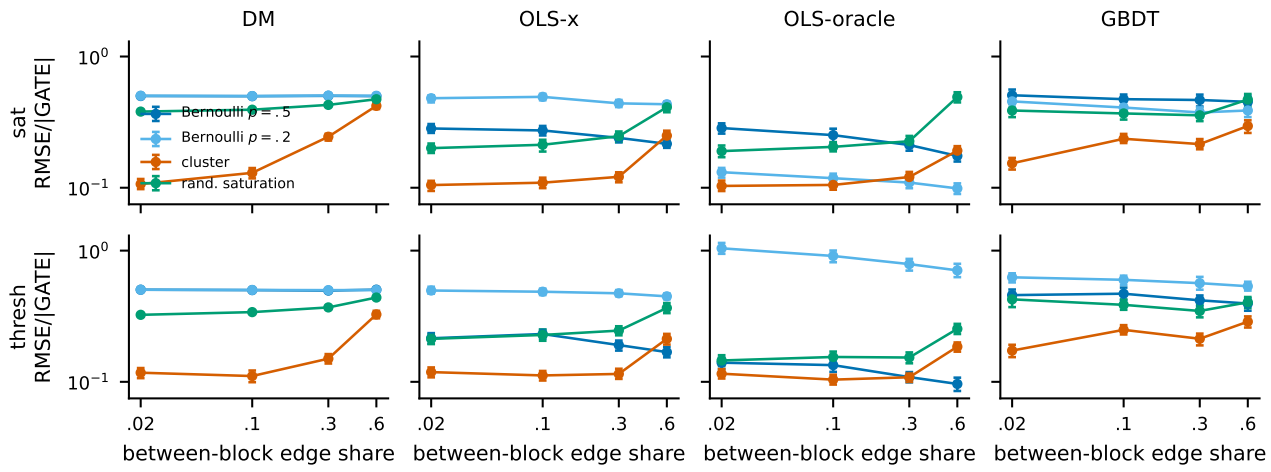


Figure 4. Design comparison on SBM graphs when the between-block edge share grows (n=3000, 200 replicates, 95% bootstrap CIs). Rows: mechanism; columns: estimator. Relative RMSE (RMSE/—GATE—).

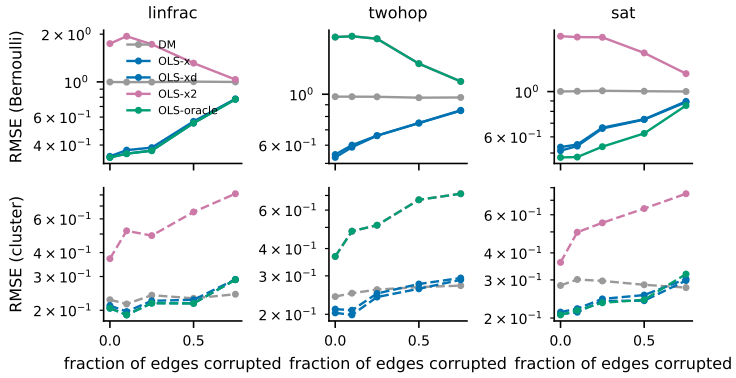


Figure 5. RMSE when the analyst's graph is corrupted (fraction of true edges replaced by random ones), SBM, 200 replicates; top: Bernoulli, bottom: cluster randomisation. Oracle uses the true exposure form on the corrupted graph.

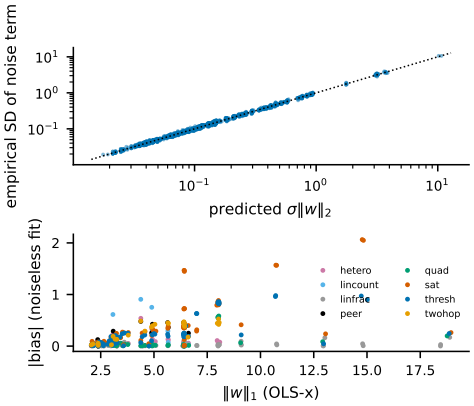


Figure 6. Top: empirical SD of the noise term vs. $\sigma\|w\|_2$ (all OLS configurations). Bottom: |bias| of OLS-x vs. $\|w\|_1$ (correct graph).

6. Discussion, limitations and scope

What the evidence supports. In our simulations a linear regression of outcomes on own treatment and treated-neighbour fraction is the most reliable regression GATE estimator across mechanisms: its noise error is exactly $\sigma\|w\|_2$ and scales as $\sqrt{d/n}$ (R2); its bias is governed by how curved the exposure–response is *near the exposure values the design generates* (R3), so it is small under cluster designs with good clusters and under linear-in-exposure interference, and large under Bernoulli designs with curved responses. Replacing the linear model by flexible learners or by many features did not help (R5); neither did the correct model form when the design leaves the nonlinear region unexcited (R4). These results suggest that the practical lever is the *design* (clusters, treatment probability, saturation) together with realised-assignment diagnostics $\|w\|_1, \|w\|_2$, rather than model sophistication.

Practical guidance (conditional on our simulations). (i) Compute $\|w\|_2$ and $\|w\|_1$ from the realised assignment; $\hat{\sigma}\|w\|_2$ is a reliable noise-error scale, and a large $\|w\|_1$ with a curved response flags extrapolation risk. (ii) At high degree relative to n (d/n large) and no good clusters, regression may be worse than DM; interference needs to be a substantial share of the GATE (here $\gtrsim 25\%$) for regression to pay off. (iii) Use cluster randomisation only if the graph has a small edge cut for the clusters; otherwise its variance cost is not repaid. (iv) Use cluster-robust SEs on the randomisation clusters rather than i.i.d./HC SEs, and treat intervals as optimistic when bias is plausible. (v) Be sceptical of the graph: corrupted edges bias Bernoulli-design estimates roughly proportionally to the corruption.

Limitations. (1) *Synthetic outcomes*: the mechanisms

Table 2. Error anatomy of OLS-x on SBM (n=3000, 10% between-block edges): bias, SD of the assignment-induced part S, SD of the noise part N, RMSE, and realised weight norms (200 replicates).

Mech.	Design	bias	SD_S	SD_N	RMSE	$\|w\|_1$	$\|w\|_2$
sat	Bernoulli p=.5	-0.39	0.35	0.15	0.55	6.4	0.147
sat	Bernoulli p=.2	+0.88	0.42	0.18	0.99	8.0	0.184
sat	cluster	+0.01	0.21	0.05	0.22	2.4	0.049
sat	rand. saturation	-0.19	0.38	0.07	0.43	3.2	0.067
thresh	Bernoulli p=.5	+0.25	0.38	0.13	0.46	6.4	0.147
thresh	Bernoulli p=.2	-0.85	0.43	0.18	0.97	8.1	0.185
thresh	cluster	-0.00	0.22	0.05	0.22	2.4	0.049
thresh	rand. saturation	+0.21	0.40	0.07	0.46	3.2	0.068
hetero	Bernoulli p=.5	-0.04	0.35	0.14	0.37	6.4	0.147
hetero	Bernoulli p=.2	-0.07	0.43	0.19	0.46	8.0	0.184
hetero	cluster	+0.00	0.23	0.05	0.24	2.4	0.049
hetero	rand. saturation	-0.01	0.36	0.06	0.36	3.2	0.067

Table 3. Coverage of nominal 95% intervals for the GATE (OLS-x, n=3000, 500 replicates) with i.i.d., HC3 and design-cluster-robust (CR) SEs; last two columns: mean SE/true SD of the estimator.

Graph	Mech.	Des.	bias	RMSE	iid	HC3	CR	$\overline{\text{SE}}_{iid}$/SD	$\overline{\text{SE}}_{CR}$/SD
SBM	linfrac	Bern	+0.01	0.36	0.76	0.76	0.93	0.59	0.97
SBM	linfrac	Clust	+0.00	0.22	0.48	0.48	0.92	0.32	0.93
SBM	sat	Bern	-0.35	0.50	0.55	0.56	0.82	0.58	0.98
SBM	sat	Clust	-0.02	0.23	0.45	0.46	0.90	0.31	0.91
ER	linfrac	Bern	+0.01	0.27	0.91	0.91	0.89	0.84	0.84
ER	linfrac	Clust	+0.01	0.32	0.79	0.79	0.82	0.64	0.69
ER	sat	Bern	-0.37	0.47	0.61	0.63	0.59	0.80	0.81
ER	sat	Clust	-0.35	0.46	0.56	0.57	0.60	0.68	0.74

(eight, all of "one-hop function of exposure" type except `twohop`/`peer`) are our own choices, and all conclusions are conditional on them; real interference may differ. Absolute error levels depend on our baseline heterogeneity and noise scale (the assignment-induced variance from baselines dominated the noise variance), and on the chosen normalisation (GATE $= 2$). (2) *One population per configuration*: graphs, baselines and community effects are drawn once (seed 0) and the replicates vary only assignment and noise; variability across graph draws is not quantified, and the 6 graphs are few. The reported CIs thus reflect Monte-Carlo, not population, uncertainty. (3) The error analysis (Prop. 3.1) is exact for OLS, but Prop. 3.2 is an asymptotic statement for regular graphs with independent neighbour treatments; we verified it numerically with a finite-d binomial version (all 54 comparisons within 2.5 s.e., two beyond 1.96; at $\bar{d} \geq 64$ the MC error is large), and it does not cover heavy-tailed degrees (BA, Stanford3), where the relevant d is not obvious. (4) *Estimators not studied*: we did not implement the full methods of Chin (2019) (resampling inference) or Han & Ugander (2023) (covariate selection and block bootstrap), graph neural networks (Kipf & Welling, 2017), staggered rollouts (Cortez et al., 2022), or covariate-adjusted/regularised variants that could reduce the variance floor or the cluster-level baseline variance; GBDT was used with one fixed hyper-parameter setting. The Hájek baseline uses a single exposure threshold (q=0.75) and thus targets a different estimand. (5) We use one clustering algorithm (Louvain at a hand-picked resolution per graph, chosen once before the main runs; Exp. 10 studies the resolution) and equal-probability cluster-level assignment. (6) Exp. 6 and 3 use different SBM block counts (K=15 and 30); comparisons across them should not be made. (7) Replicate counts (100–500) limit the resolution of small biases ($\lesssim 0.03$). (8) Compute: all experiments used 4 CPU cores; no hyper-parameter tuning was done on test outcomes.

Negative and unexpected results. (a) Regression adds little over DM under cluster randomisation with good clusters (and DM is best on ego-Facebook), so the accuracy gain is mostly due to the design. (b) Flexible learners were worse than the simple linear model. (c) The oracle model was far from the best in some conditions (R4). (d) Standard SEs are badly miscalibrated. (e) $\|w\|_1$ is a poor general predictor of bias (R9).

Impact Statement

This paper studies statistical methodology for online experiments using synthetic outcomes and public, anonymised graphs; we see no specific negative societal consequences beyond those of A/B testing in general. Mis-estimating spillovers can lead to wrong product decisions, which is why we emphasise error bounds and failure conditions.

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A. Reproduction map and environment

All code is in `code/` (library `gate/`); see `README.md` for commands. Python 3.11.2, numpy 2.4.6, scipy 1.17.1, networkx 3.6.1, pandas 3.1.0, matplotlib 3.11.2, scikit-learn 1.9.1. Graphs: SNAP ego-Facebook (`facebook_combined.txt.gz`, sha256 `125e84db...ad035d7`, 4039 nodes, 88,234 edges; McAuley & Leskovec 2012) and Network Repository `socfb-Stanford3` (zip sha256 `b1221e41...5e8720d`; 11,586 nodes, 568,309 edges, CC BY-SA; Rossi & Ahmed 2015; Traud et al. 2012); both downloaded on 2026-11-01 and checked against the hashes. The largest connected component is used (both graphs are connected). Population seed 0; replicate r of a configuration uses seed $[2026, \mathrm{crc32}(\mathrm{config}), r]$. Machine: 4 usable CPU cores. Wall-clock: Exp. 1 36 s, Exp. 2 25 min, Exp. 3 22 min, Exp. 4 4.3 min, Exp. 5 13 min, Exp. 6 20 min, Exp. 7 9 s, Exp. 8 13 s, Exp. 9 7 min, Exp. 10 13 min, Exp. 11 23 s.

Table 4. Experiment → script → artifact (all scripts in `code/`; run `python experiments.py <exp>` first).

Exp. (name)	Script	Artifacts
1 (`e1_var`)	`fig_e1.py`	Fig. 1
2 (`e2_main`)	`fig_e2.py`, `tab_e2.py`	Table 1, Fig. 7
3 (`e3_design`)	`fig_e3.py`	Fig. 4, Table 2
4 (`e4_scale`)	`fig_e4.py`	Fig. 9
5 (`e5_dosenoise`)	`fig_e5.py`	Fig. 10
6 (`e6_degree`)	`fig_e6.py`	Fig. 3
7 (`e7_infer`)	`tab_e7.py`	Tables 3, 8
8 (`e8_misspec_graph`)	`fig_e8.py`	Fig. 5
9 (`e9_prob`)	`fig_e9.py`	Fig. 2
10 (`e10_clusterres`)	`fig_e10.py`, `tab_app.py`	Table 6
11 (`e11_homophily`)	`tab_app.py`	Table 7
1–11	`fig_diag.py`	Fig. 6, R2/R9 numbers
6, 9	`stats_theory.py`, `tab_theory.py`	Table 5, R3 numbers

B. Implementation details

Estimators. OLS uses a tiny ridge ($10^{-8}n$ on the diagonal) for numerical stability only. Feature sets: x $= (z, x)$; c $= (z, \mathrm{count}/\bar{d})$; xd $= (z, x, \mathrm{zlogdeg})$; cd $= (z, \mathrm{count}/\bar{d}, \mathrm{zlogdeg})$; x2 $= (z, x, Wx, \mathrm{zlogdeg})$; quad $= (z, x, x^2, zx, \mathrm{zlogdeg})$; rich $= (z, x, x^2, x^3, zx, \mathrm{zlogdeg}, z\,\mathrm{zlogdeg}, x\,\mathrm{zlogdeg}, Wx)$; oracle uses the true exposure basis (plus zlogdeg; for `hetero` it also uses the latent u, for `peer` the exact $(I - \rho W)^{-1} z$). Note that OLS-x omits the degree covariate, while the baseline contains a degree term. All regressions predict at $z{=}\mathbf{1}$ and $z{=}\mathbf{0}$ by recomputing the features on the all-treated and all-control vectors. GBDT: scikit-learn `HistGradientBoostingRegressor` (150 iterations, learning rate 0.1, max depth 4, min leaf 20) on $(z, x, \mathrm{count}/\bar{d}, Wx, \mathrm{zlogdeg})$, not tuned. Hájek: exposure indicators $\{z_i{=}1, x_i \geq 0.75\}$ and $\{z_i{=}0, x_i \leq 0.25\}$, probabilities by 400 Monte-Carlo design draws (floored at $1/401$), Hájek-normalised; undefined when either group

Table 5. Prop. 3.2 check, SBM graph, Bernoulli(p): empirical bias of the noiseless OLS-x fit (± Monte-Carlo s.e.) vs. the finite-degree prediction using the realised degree sequence (Exp. 9, 200 replicates).

Mech.	p	empirical	s.e.	predicted	diff
linfrac	0.05	-0.007	0.055	+0.000	-0.007
linfrac	0.10	+0.023	0.040	-0.000	+0.023
linfrac	0.20	+0.016	0.031	-0.000	+0.016
linfrac	0.30	-0.010	0.026	+0.000	-0.010
linfrac	0.50	+0.029	0.024	-0.000	+0.029
sat	0.05	+2.049	0.059	+2.016	+0.033
sat	0.10	+1.566	0.041	+1.556	+0.009
sat	0.20	+0.858	0.030	+0.826	+0.032
sat	0.30	+0.291	0.026	+0.293	-0.002
sat	0.50	-0.365	0.023	-0.369	+0.004
thresh	0.05	-0.901	0.055	-0.973	+0.072
thresh	0.10	-0.960	0.041	-0.950	-0.011
thresh	0.20	-0.832	0.030	-0.852	+0.020
thresh	0.30	-0.630	0.024	-0.642	+0.012
thresh	0.50	+0.248	0.023	+0.231	+0.018

is empty, counted as failure (and excluded from RMSE; failure rates are in the summary CSVs).

Mechanism parameters. τ_d=1, γ=1, `sat`: κ=4; `thresh`: centre 0.7, slope 10; `peer`: ρ=0.5; `hetero`: $(1 + 0.5u_i)$ with $u = 0.6\,\mathrm{zlogdeg} + 0.8\,\xi$ standardised; baseline $a_i = 1 + 0.5\,\mathrm{zlogdeg}_i + s_b\, b_{c(i)} + 0.7\nu_i$, s_b=0.5 ($c(i)$: Louvain community at resolution 1). The GATE is computed exactly; it equals 2.000 for all mechanisms except `hetero`/`peer` where it is the computed value (also ≈ 2). The "noiseless refit" gives the assignment-induced part S.

Clusters. Louvain (Blondel et al., 2008) via networkx with seed 0; resolution per graph for the main grid: ER 3 (K=110, cut 0.82), BA 3 (111, 0.82), WS 3 (49, 0.16), SBM 1 (30, 0.10), egoFB 3 (27, 0.07), Stanford3 3 (57, 0.64), chosen once after inspecting K and cut and before looking at any outcome. Resolution 1 on ER/BA gives $K \approx$ 10–12; we did not tune further. Cluster assignment treats $\mathrm{round}(pK)$ clusters.

Graph corruption (Exp. 8). A fraction r of edges is deleted uniformly and the same number of uniformly random pairs is added; features are computed on this graph, outcomes on the true graph.

Inference (Exp. 7). With $e = y - X\hat{\beta}$: HC0 = $\sqrt{\sum w_i^2 e_i^2}$, HC3 uses $e_i/(1 - h_{ii})$, CR = $\sqrt{\sum_g (\sum_{i\in g} w_i e_i)^2}$ over design clusters (Louvain, same as the cluster design; for Bernoulli designs the same partition is used only for the SE), i.i.d. = $\hat{\sigma}\|w\|_2$; intervals $\hat{\tau} \pm 1.96\,\mathrm{SE}$ vs. the exact GATE.

C. Additional results

D. Process log, access statement and checklist

Process notes (honest account). (i) A first launch used 38 worker processes although only 4 cores were available (cgroup limit); it was killed and restarted with 4 workers (no results from it were used). (ii) Exp. 6 first crashed for $\bar{d}$=128 on the 30-block SBM (edge probability above 1); it was re-run with 15 blocks of 200 nodes, and the corrected configuration is the one reported. (iii) Exp. 10 and 11 were added after the first results to address cluster granularity and homophily. (iv) An exploratory smoke test (`e0_smoke`, 20 replicates) preceded the main runs and was not used. (v) Design choices that were made after seeing smoke-test output: dropping the count-feature estimator `c` from the headline comparison (it is confounded with degree in our baseline), adding the community baseline-effect ablation (Exp. 1, 11). (vi) No hyper-parameter was tuned on outcomes. Nothing in the paper comes from a run that is not reproducible with the shipped code, and every reported number is regenerated by the scripts above.

Restricted content. The task excluded one published paper and its derivatives. We did not search for it; our literature searches were by the exact titles of well-known papers (listed in the references) and the starter list. We did not knowingly encounter any excluded material.

Checklist. Claims match evidence: yes (Sec. 5; limits in Sec. 6). Theory assumptions and proofs: Props. 3.1–3.2 state assumptions; Prop. 3.2 is a sketch. Code/data: provided with hashes and commands. Statistical significance: bootstrap CIs and Monte-Carlo s.e. (replicates 100–500; one fixed population per configuration). Compute: 4 CPU cores, ≈ 1.9 h of simulation wall-clock in total. Licences: both graphs are public (CC BY-SA / cited). No human subjects. Impact: see Impact Statement.

Table 6. Cluster granularity (Exp. 10): RMSE for `sat` (left four) and `thresh` (right four). Res.: Louvain resolution; K clusters; cut: fraction of edges between clusters; cluster design with half of the clusters treated; 200 replicates.

Graph	Res.	K	cut	DM	OLS-x	Orac.	GBDT	DM	OLS-x	Orac.	GBDT
egoFB	1	16	0.04	0.34	0.34	0.37	0.97	0.34	0.35	0.36	0.95
egoFB	3	27	0.07	0.36	0.41	0.41	0.99	0.33	0.39	0.37	1.06
egoFB	10	238	0.53	0.40	0.26	0.29	1.13	0.29	0.26	0.23	1.12
egoFB	30	672	0.73	0.56	0.31	0.29	1.06	0.45	0.30	0.22	1.12
Stanford3	1	10	0.46	0.77	0.60	0.66	0.61	0.68	0.60	0.61	0.56
Stanford3	3	57	0.64	0.66	0.49	0.57	0.54	0.53	0.64	0.53	0.54
Stanford3	10	197	0.74	0.70	0.41	0.52	0.47	0.58	0.58	0.44	0.46
WS	1	27	0.13	0.32	0.24	0.23	0.62	0.22	0.22	0.21	0.65
WS	3	49	0.16	0.28	0.19	0.18	0.67	0.21	0.20	0.18	0.66
WS	10	89	0.21	0.33	0.20	0.18	0.65	0.20	0.18	0.15	0.57
WS	30	156	0.30	0.41	0.17	0.15	0.54	0.25	0.17	0.12	0.60
SBM(0.3)	1	30	0.31	0.50	0.23	0.22	0.42	0.33	0.24	0.23	0.41
SBM(0.3)	3	30	0.31	0.50	0.23	0.22	0.41	0.30	0.23	0.21	0.41
SBM(0.3)	30	376	0.78	0.86	0.49	0.45	0.76	0.73	0.48	0.28	0.66

Table 7. Community-level baseline effect sd $s_b \in \{0, 0.5, 1, 2\}$ (Exp. 11, `sat`): RMSE (GATE $= 2$), 300 replicates.

Graph	Des.	Est.	0	0.5	1	2
SBM	Bern	DM	1.00	1.00	1.00	1.01
SBM	Bern	OLS-x	0.41	0.51	0.74	1.37
SBM	Bern	OLS-oracle	0.25	0.52	0.89	1.77
SBM	Clust	DM	0.18	0.28	0.47	0.87
SBM	Clust	OLS-x	0.06	0.21	0.41	0.84
SBM	Clust	OLS-oracle	0.05	0.20	0.41	0.85
WS	Bern	DM	1.00	1.00	1.00	1.00
WS	Bern	OLS-x	0.42	0.55	0.73	1.41
WS	Bern	OLS-oracle	0.27	0.53	0.94	1.92
WS	Clust	DM	0.24	0.28	0.38	0.69
WS	Clust	OLS-x	0.07	0.19	0.37	0.78
WS	Clust	OLS-oracle	0.05	0.18	0.35	0.75
egoFB	Bern	DM	1.01	1.00	1.00	1.00
egoFB	Bern	OLS-x	0.66	0.62	0.90	1.63
egoFB	Bern	OLS-oracle	0.15	0.45	0.85	1.61
egoFB	Clust	DM	0.22	0.33	0.64	1.37
egoFB	Clust	OLS-x	0.22	0.37	0.75	1.63
egoFB	Clust	OLS-oracle	0.04	0.37	0.77	1.60

Table 8. Full coverage table (Exp. 7; columns as in Table 3).

Graph	Mech.	Des.	bias	RMSE	iid	HC3	CR	$\overline{\text{SE}}_{iid}$/SD	$\overline{\text{SE}}_{CR}$/SD
SBM	hetero	Bern	-0.05	0.37	0.74	0.76	0.94	0.61	1.01
SBM	hetero	Clust	-0.02	0.22	0.48	0.48	0.93	0.34	0.97
SBM	linfrac	Bern	+0.01	0.36	0.76	0.76	0.93	0.59	0.97
SBM	linfrac	Clust	+0.00	0.22	0.48	0.48	0.92	0.32	0.93
SBM	peer	Bern	-0.41	0.76	0.47	0.47	0.87	0.39	0.93
SBM	peer	Clust	-0.10	0.40	0.34	0.34	0.94	0.22	0.99
SBM	sat	Bern	-0.35	0.50	0.55	0.56	0.82	0.58	0.98
SBM	sat	Clust	-0.02	0.23	0.45	0.46	0.90	0.31	0.91
ER	hetero	Bern	-0.07	0.28	0.92	0.93	0.88	0.88	0.89
ER	hetero	Clust	-0.07	0.32	0.80	0.80	0.82	0.67	0.73
ER	linfrac	Bern	+0.01	0.27	0.91	0.91	0.89	0.84	0.84
ER	linfrac	Clust	+0.01	0.32	0.79	0.79	0.82	0.64	0.69
ER	peer	Bern	-0.47	0.57	0.49	0.50	0.53	0.75	0.78
ER	peer	Clust	-0.41	0.57	0.48	0.48	0.54	0.54	0.60
ER	sat	Bern	-0.37	0.47	0.61	0.63	0.59	0.80	0.81
ER	sat	Clust	-0.35	0.46	0.56	0.57	0.60	0.68	0.74

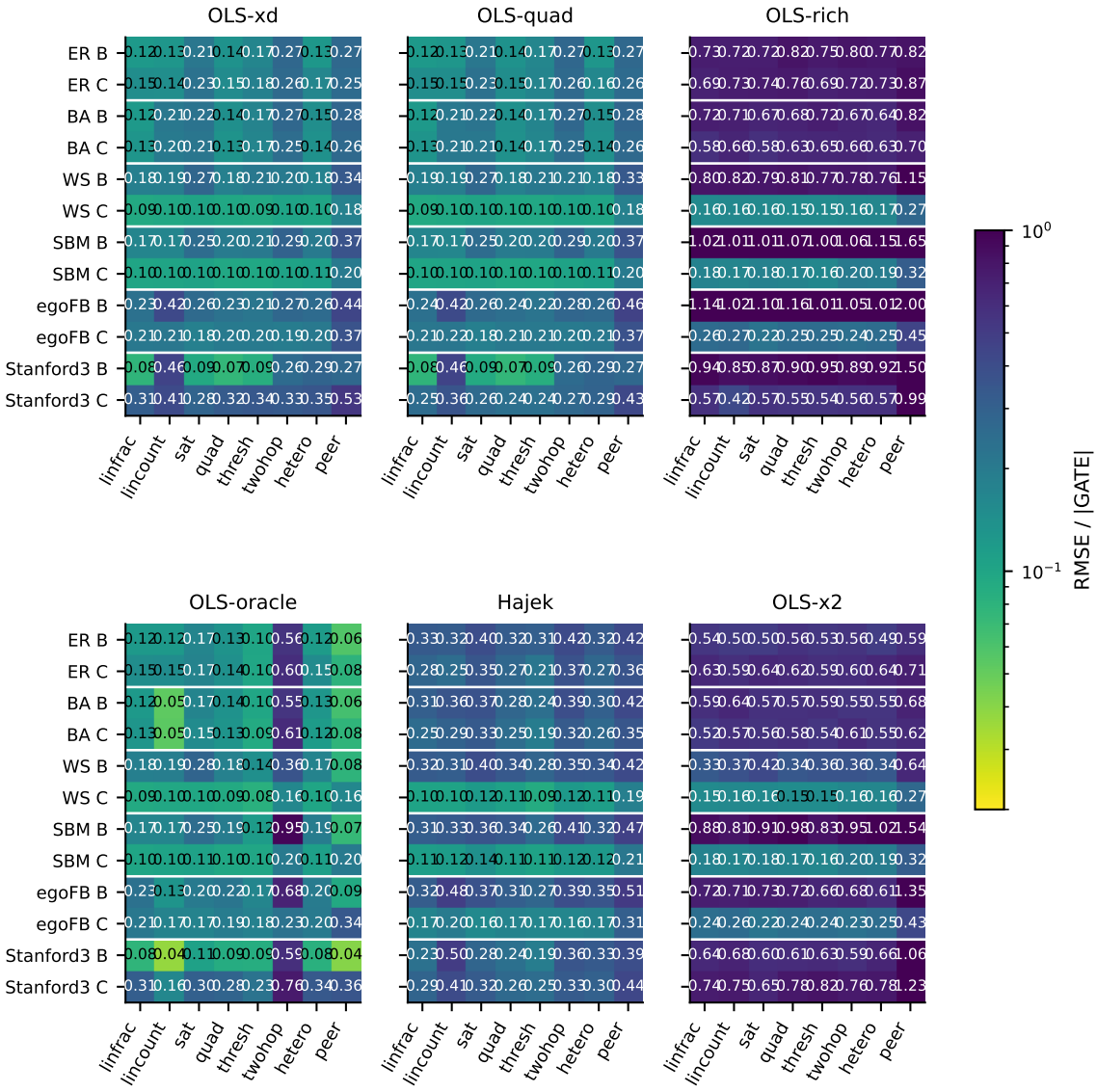


Figure 7. Relative RMSE of the other estimators in the main grid (Exp. 2): rows are graph and design (B: Bernoulli, C: cluster), columns mechanisms. 200 replicates per cell.

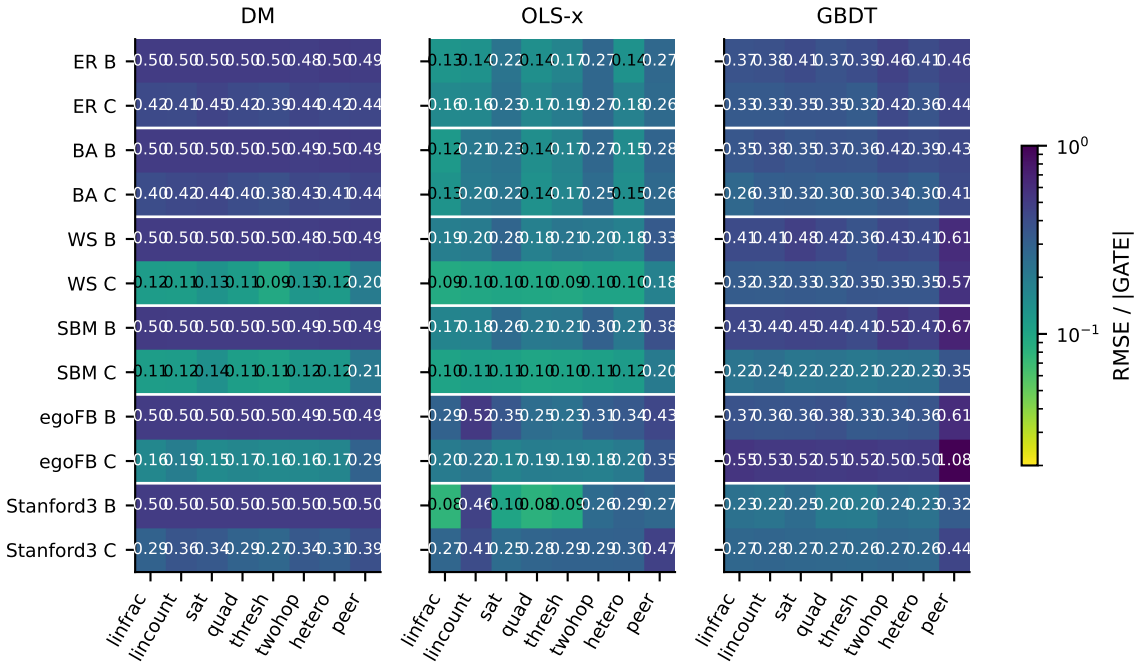


Figure 8. Heatmap version of the main grid (DM, OLS-x, GBDT); relative RMSE.

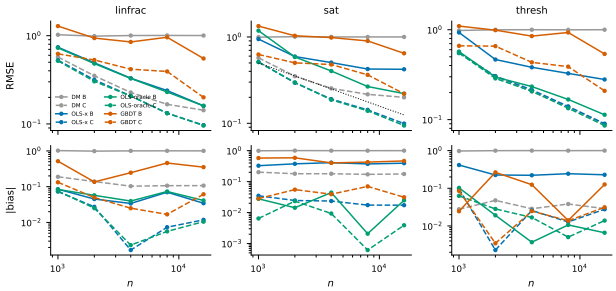


Figure 9. Scaling with n (SBM, mean degree 16, 100 nodes per block, 100 replicates). Solid: Bernoulli, dashed: cluster. Top: RMSE; bottom: |bias|. Dotted: slope $-1/2$. The GBDT bias and the OLS-x bias for `sat`/`thresh` under Bernoulli do not vanish with n.

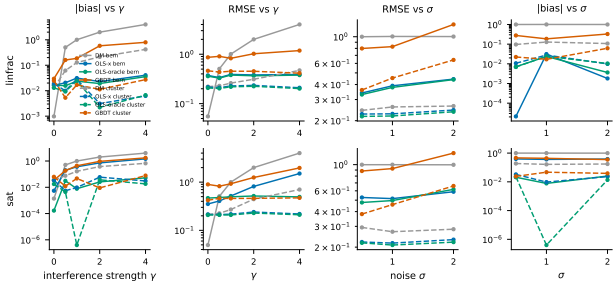


Figure 10. Interference strength γ and noise σ (SBM, 200 replicates). Solid: Bernoulli; dashed: cluster. Rows: mechanism.